

Visible Learning and whole language: revisiting the 'garbage in, garbage out' problem

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ABSTRACT

A decade after the publication of Visible Learning: A Synthesis of Over 800 Meta-Analyses Relating to Achievement, the influence of John Hattie's statistical work continues to grow. This paper critically examines one aspect of Hattie's 'synthesis' of education meta-analyses, his summation of research data for whole language literacy teaching. Of the four meta-analyses reported as being on whole language in Visible Learning, one is in fact a survey of research on language experience. Hattie uncritically accepted another meta-analysis, which reported that whole language triggers dramatic student unlearning, despite ample reasons to question its findings. On the other hand Hattie, revisiting the sole meta-analysis, reported finding a positive effect size for whole language in an unusual manner, with selected research studies removed to produce an effect size of zero. This paper argues that a prior bias on the part of the author of Visible Learning – against whole language and in favour of synthetic phonics and direct instruction – likely explains this interpretation of the research data on whole language. Potential implications for basing pedagogical practice and public policy decisions on aggregated effect sizes from meta-analyses are discussed.

Introduction

The reception of John Hattie's (2009) *Visible Learning: A synthesis of over 800 meta-analyses relating to achievement* saw a *Times Educational Supplement* reviewer praise the author for having found 'education's equivalent to the search for the Holy Grail' (Mansell, 2008). In Australia, Hattie's recommendations for raising student achievement are now informing different states' education departments' policies. Those of us within the Victorian public education system are now advised to implement ten 'High Impact Teaching Strategies', with Hattie's *Visible Learning* listed as the primary reference for each one (Department of Education and Training Victoria, 2017a), and utilise a new Literacy Toolkit (Department of Education and Training Victoria, 2017b) that prominently references Hattie's work and a related *Visible Learning for Literacy* (Fisher et al., 2016) book published by Corwin, owner of the Visible Learning^{plus} brand.

The 'visible learning' mantra has been taken up internationally, perhaps unsurprisingly given how well it fits within what Sahlberg (2012) identified as the five central features of the Global Education Reform Movement (GERM): standardised curricula and assessments, a one-sided focus on literacy and numeracy, the search for low-risk ways to reach learning goals, the use of corporate management models to drive improvement, and finally the adoption of test-based 'accountability' policies. Hattie's work has, however, come under growing scrutiny and criticism from mathematicians and statisticians: for an extensive summary see Lilley (2016). Most sharply, Bergeron (2017, p. 238) characterised 'Hattie's methodology as pseudoscience ... When taking the necessary in-depth look at *Visible Learning* with the eye of an expert, we find not a mighty castle but a fragile house of cards that quickly falls apart'.

Hattie (n.d.) has responded to earlier challenges to his statistics by conceding certain methodological mistakes,

but stating: 'I regret these errors, although NONE of them reduce the impact of the story that underlies the VL [*Visible Learning*] messages.' In *Visible Learning*, Hattie noted (2009, p. 237): 'The current obsession with evidence-based too often ignores the lens that researchers use to make decisions about what to include (as evidence), what to exclude, and how to marshal the evidence to tell their story. It is the *story* that is meant to be the compelling contribution – it is my lens on this evidence.'

As Hattie's statistics have come under greater scrutiny, his emphasis on the narrative value of *Visible Learning* has increased. He and Hamilton (2018, p. 46) insisted that 'the key here is that it is the INTERPRETATION and STORY that help explain the findings that is a contender for a useful theory; not the data'.

This emphasis is peculiar given that meta-analysis emerged with the promise that it offered, in the words of Gene Glass (1976, p. 3), who coined the term, 'a rigorous alternative to the casual, narrative discussions of research studies'. Hattie's remarks raise the question as to what exactly *is* the connection between the *Visible Learning* 'story' and education research data. Through a close reading of Hattie's assessment of the effectiveness of whole language teaching, this article will argue that there is in fact no scientifically defensible connection, at least in relation to one crucial area of literacy research. The *Visible Learning* 'story' is one explicitly hostile to constructivist and progressive pedagogies, and in favour of direct instruction and related conservative pedagogies. It will be argued that this 'story' did not emerge out of the research data, but rather reflected prior bias that drove a series of questionable decisions about what to include and exclude as relevant literacy research, contributing to an erroneous evaluation of the data. The end result is a demonstrably distorted portrait of the evidence base for whole language. More broadly called into question is the legitimacy of attempting to base either pedagogical practice or public policy initiatives on aggregated meta-analyses, as presented in *Visible Learning*.

Hattie's assessment of whole language

Hattie's method involves the aggregation of education research meta-analyses, that is, statistical papers that combine the results of multiple research studies to arrive at an 'effect size' for a given intervention, method, or factor affecting student learning. An effect size (symbolised with '*d*') of 1.0 indicates an increase of one standard deviation for school achievement, 'which is typically associated with advancing children's achievement by two to three years, improving the rate of learning by 50%' (2009, p. 7). Hattie found that

the average effect size across all the meta-analyses he examined to be $d = 0.4$. This he proposed as the benchmark or 'hinge point' for effective teaching intervention, as it 'sets a level where the effects of innovation enhance achievement in such a way that we can notice real-world differences, and this should become a benchmark of such real-world change' (Hattie, 2009, p. 17).

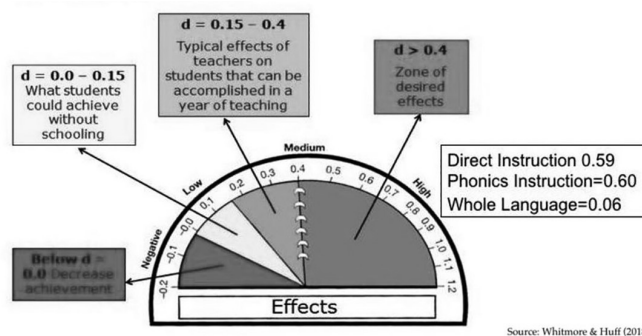


Figure 1. Hattie's barometers of influence (Whitmore and Huff 2018, slide 19)

Within this framework, Hattie concluded that whole language had an effect size of just 0.06. Of the 25 different factors categorised as 'curricula' by Hattie (covering, for example, repeated reading programs and integrated curriculum programs), whole language was found to be the *least* effective.

In effect, according to this calculation, whole language teaching has zero real impact on student learning. Hattie explained: 'The zone between $d = 0.0$ and $d = 0.15$ is what students could probably achieve if there was no schooling (and is estimated from the findings in countries with no or limited schooling)' (2009, p. 20). He specifically added: 'In summary, whole language programs have negligible effects on learning to read – be it on word recognition or on comprehension.' (Hattie, 2009, p. 138).

If correct, this would be a highly significant finding. A student in a classroom with a whole language literacy teacher would learn to read just as effectively if they had never set foot inside a school. Hattie, it appears, has definitively resolved the decades-old, highly contested debate between proponents of systematic synthetic phonics and advocates of holistic, literature-based reading and writing teaching, in favour of the former. Phonics-based programs were assigned an effect size of 0.60 in *Visible Learning*, with Hattie (2009) concluding, 'Overall, phonics instruction is powerful in the process of learning to read – both for reading skills and for reading comprehension' (p. 134).

Several figures favouring different phonics programs have enthusiastically welcomed Hattie's findings. Noel Pearson, director of the Cape York Institute for Policy and Leadership and a supporter of the MultiLit

synthetic phonics program, said for example, 'The reading wars should be over – the evidence in favour of phonics instruction (0.60) v whole language (0.06) is stark' (Pearson, 2011).

'Garbage in – garbage out'

Before we can agree that proponents of synthetic phonics have 'won' the so-called reading wars – the politicised public debate over literacy instruction, fuelled by recurring media-driven moral panics over purported declining standards (Gannon & Sawyer, 2007) – it is necessary to examine Hattie's method, and the research underlying the method.

Both meta-analysis and aggregations of meta-analyses, which Hattie and others have described as 'meta-meta-analysis', have been challenged on different grounds. H.J. Eysenck (1978) famously described meta-analysis as an 'exercise in mega-silliness'. He criticised the indiscriminate nature of meta-analysis, with every research study related to a selected area used to calculate an effect size, regardless of the different studies' quality or rigour. Eysenck (1978) wrote: 'A mass of reports – good, bad, and indifferent – are fed into the computer in the hope that people will cease caring about the quality of the material on which the conclusions are based. [...] "Garbage in – garbage out" is a well-known axiom of computer specialists; it applies here with equal force' (p. 517).

More recently, Pawson (2006) argued along similar lines as he proposed a 'realist synthesis' to understand different areas of research as an alternative to meta-analysis:

At every stage of the meta-analytic review, simplifications are made. Hypotheses are abridged, studies are dropped, program details are filtered out, contextual information is eliminated, selected findings are utilised, averages are taken, estimates are made ... such a process of simplification, standardisation and aggregation does not produce any enduring truths. It produces a descriptive summary that is likely to be disappointing or misleading or in need of further clarification. (pp. 42–43, 72)

Part of *Visible Learning*'s success is likely due to the extraordinary ambition and scope of Hattie's project. His synthesis of more than 800 education-related meta-analyses incorporates 52,000 individual research studies and as many as 236 million children. Reviewing all the different aspects of *Visible Learning*'s findings is a daunting prospect. When we focus, however, on *Visible Learning*'s treatment of whole language we are dealing with four meta-analyses, involving a total of 64 studies. Hattie's summation of this research comprises just five paragraphs, one-and-a-half pages, in *Visible*

Learning. A detailed examination of these four meta-analyses, and Hattie's interpretation of them, is both feasible and fruitful.

One aspect of the 'garbage in – garbage out' problem for meta-analyses concerns whether the various studies included in a meta-analysis have been properly selected. In assessing whether this was the case for the studies incorporated into *Visible Learning*'s assessment of whole language, the first issue is defining whole language, or at least setting parameters around what a research study into whole language looks like or does not look like. This is not a straightforward question, given that whole language has been variously described as a philosophy, an epistemology, and a movement (Wilson, 1997).

Ken Goodman (1967) – whose landmark article 'Reading: A Psycholinguistic Guessing Game', set the stage for the emergence of the whole language movement in the 1970s – suggested several guidelines for identifying what whole language is and what it is not. Whole language, he explained in a widely cited book (c.f. Watson, 1989; Dudley-Marling, 1995):

... firmly rejects such things as these: isolating skill sequences; slicing up reading and writing into grade slices; each slice neatly following and dependent on prior ones; simplifying texts by controlling their sentence structures and vocabulary, or organising them around phonic patterns; equating reading and writing with cores on tests of 'sub-skills'; [and] isolating reading and writing instruction from its use in learning, or in actual reading and writing. (Goodman, 2005, pp. 35–36)

He added:

Whole language learning builds around whole learners learning whole language in whole situations; whole language learning assumes respect for language, for the learner, and for the teacher; the focus is on meaning and not on language itself, in authentic speech and literacy events; learners are encouraged to take risks and invited to use language, in all its varieties, for their own purposes; in a whole language classroom, all the varied functions of oral and written language are appropriate and encouraged. (Goodman, 2005, p. 42)

None of the nuances in such an approach are dealt with in *Visible Learning*. Hattie opens the section on whole language (Hattie, 2009, p. 137) by relying on a meta-analysis citation:

The whole language approach to reading instruction is based on the idea that the 'acquisition of reading skills depends on the context in which these skills are presented. Individual words are learned more easily and fluently when presented within a particular context. The words gather meaning from other words around them and from the structure of the story'. (cited in Gee, 1995, p. 5)

Hattie did not quote from the rest of Gee's consideration of how to define whole language, including his recognition that 'whole language is not a canned method but rather an overall philosophy about the way children learn and the best way to teach them', and that 'many different techniques for teaching reading may fall under the auspices of whole language' (Gee, 1995, p. 6).

When a 'whole language' meta-analysis is not about whole language

Of the four whole language meta-analyses included in *Visible Learning*, the earliest published is Stahl and Miller's (1989), 'Whole language and language experience approaches for beginning reading: a quantitative research synthesis'. Hattie (2009) summarised the study's conclusions as follows:

Stahl and Miller (1989) also had zero effects ($d = 0.17$ for word recognition and $d = 0.09$ for reading comprehension) from whole language programs. Whole language approaches may be more effective in kindergarten than in first grade, they may produce higher (but still close to zero) effects on measures of word recognition than reading comprehension measures, and they are more effective when used instead of a reading readiness program. They compared the effects to basal readers, and there was a trend towards higher effects for basal reading programs. (p. 138)

There is a basic problem with this analysis – Stahl and Miller's meta-analysis conflated language experience literacy approaches with whole language teaching and learning. R.V. Allen (1964, p. 60) explained: 'Briefly stated, the language-experience approach is one which brings reading and the other communication skills together in the instructional program.' As conceptualised by the child, Allen stated (p. 59), language experience is summed up in the formula: 'What I can think about, I can talk about. What I can say I can write ... What I can write, I can read. I can read what I can write and what other people can write for me to read.' Despite noting that 'there are some important differences' between 'the earlier language experience approaches and current whole language approaches', (p. 88) Stahl and Miller insisted, 'We see these approaches as being both part of a continuous evolution, yet having some possibly important differences. Therefore, we will refer to the entire range of approaches as *whole language/ language experience approaches*' (Stahl & Miller, 1989, p. 89, original emphasis). As a consequence of this approach, the meta-analysis included multiple research studies from the early and mid-1960s, before whole language first emerged in the 1970s.

The problems involved with this become obvious as soon as the research is examined. Stahl and Miller's

meta-analysis included, for example, Brazziel and Terrell (1962), which was based on work with an experimental group of 26 African-American children and their families, mostly low income farming households in Tennessee. The researchers explained, in their four-page report: 'We wanted to learn whether a change in our approach to [school] registration, combined with parent-teacher co-operation to develop the children's readiness for learning, could overcome the effects of the heritage of a disadvantaged group of first-graders' (Brazziel & Terrell, 1962, p. 352).

The experimental registration process involved 'several innovations', including having parents meet over a six week period with their child's teacher for an hour, once a week, 'going over plans for the readiness period, studying the work of the first grade, and discussing the philosophy and purpose of the school' (Brazziel & Terrell, 1962, p. 353). The researchers also explained: 'The children were given readiness readers that provide intensified activity to develop perception, vocabulary, word reasoning, and the ability and the will to follow directions' (Brazziel & Terrell, 1962, p. 353).

The study appears to have been included in Stahl and Miller's meta-analysis because the researchers also instructed teachers to help the children construct 'experience charts' that were 'based on the children's day-to-day experiences, which they related during sharing periods' (Brazziel & Terrell, 1962, p. 354). Whether this constituted sufficient grounds for the study's inclusion in a meta-analysis on language experience is debatable. There can be no doubt, however, that Brazziel and Terrell's paper has nothing to do with whole language and should not have formed a part (however small when taken alone) of *Visible Learning's* effect size for whole language.

Several other researchers challenged Stahl; Miller's conclusions (see Schickedanz, 1990 and Edelsky, 1994). McGee and Lomax (1990, p. 135) noted, 'language experience was an important precursor to whole language, but we assert that results of a study(r)TJETg 3V5.8

Learning). In 'The Effects of Whole-Language Instruction: An Update and a Reappraisal', Stahl, McKenna, and Pagnucco (1994) recognised the distinction between 'whole language and the earlier language experience approaches', with the 'beginning of the whole language movement' traced to the year 1976 (Stahl et al., 1994, p. 175). This meta-analysis 'did not include any studies dealing purely with language experience approaches [because] at this time, such a label would be anachronistic' (Stahl et al., 1994, p. 176). Equating language experience with whole language was acknowledged as 'anachronistic' in 1994 – yet more than a decade later Hattie included in *Visible Learning* a meta-analysis doing just that, without qualification or explanation.

When a whole language meta-analysis finds children's learning going backwards

The overall effect size for whole language, $d = 0.06$, was calculated in *Visible Learning* from a mean calculation of the effect sizes from the four meta-analyses he examined.

For the first of these four reports, which we have seen ought never to have been included in a review of whole language research studies, Hattie calculated an effect size of 0.09. For the other three, the effect sizes are 0.15 for Stahl's follow-up meta-analysis (Stahl, McKenna & Pagnucco, 1994), 0.65 for Gee's study (1995), and -0.65 for Jaynes and Littell (2000). No weightings are considered in *Visible Learning* for different factors underlying each meta-analysis, such as the number or quality of the various studies within each meta-analysis. Hattie instead simply added 0.09, 0.15, 0.65, and -0.65 and divided the total by 4.

Reporting an average $d = 0.06$ for whole language,

when the two most recent meta-analyses have diametrically opposed findings (0.65 and -0.65), is clearly problematic. It brings to mind the joke about the man with his head in the oven and feet in the fridge – the statistician declares his average temperature perfectly fine. Hattie (2009) acknowledges in *Visible Learning*: 'At first look there is remarkable divergence in the overall effects from the four meta-analyses on whole language approaches' (p. 137).

Let us examine how Hattie accounts for this 'remarkable divergence' by first looking at the most striking effect size reported for whole language, *negative* 0.65 in the Jaynes and Littell (2000) study. This is an extreme outlier in *Visible Learning* – of the more than 800 meta-analyses, just *two* found an effect size worse than -0.5. If Jaynes and Littell are correct, then whole language causes dramatic student *unlearning*.

This is an extraordinary conclusion for any meta-analysis concerned with student learning. Hattie, we will recall, defined a positive 0.4 effect size as the 'hinge point', because, 'When teachers claim that they are having a positive effect on achievement or when a policy improves achievement this is almost a trivial claim: virtually everything works. One only needs a pulse and we can improve achievement' (2009, p. 16).

The Jaynes and Littell meta-analysis suggests that a whole language teacher would do less damage if he or she did *not* have a pulse. Extraordinary claims require extraordinary evidence – before accepting Jaynes and Littell's $d = -0.65$ finding, to what extent did Hattie critically review the meta-analysis, including its selection of research studies and its effect size calculations? There is no indication in *Visible Learning* that any scholarly scrutiny was brought to bear. The one

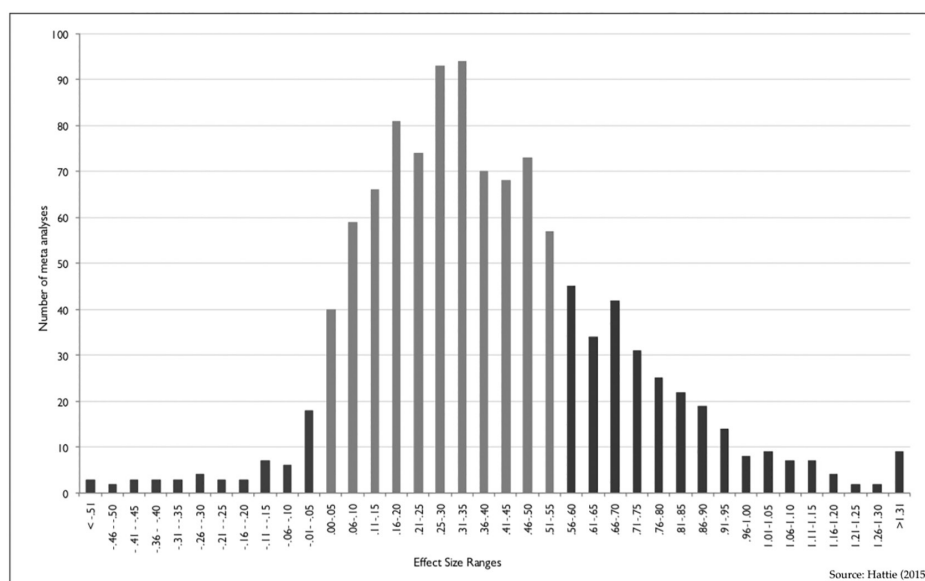


Figure 2. Relative ranking of a selection of popular education 'fixes', by effect size.

paragraph referring to the meta-analysis simply reports:

The most recent study found a substantial negative relationship between whole language interventions compared to basal readers and learning to read programs. Jeynes and Littell (2000) investigated the effect of whole language instruction on the literacy of low-socioeconomic status students from kindergarten through to grade level 3 and found that low socioeconomic status children receiving basal readers did consistently better than their counterparts receiving whole language instruction.' (p. 138)

Behind the -0.65 effect size

The conclusions of the Jeynes and Littell (2000) meta-analysis are in fact much more ambiguous than *Visible Learning's* summary suggests. The researchers distinguished between four classifications of whole language studies – pure, specific, broad, and eclectic. These are categories that find no reflection in any other literature on whole language. Jeynes and Littell nevertheless defined 'pure' whole language studies as those 'that whole language enthusiasts would agree represented the best features of whole language', with three conditions attached: '(1) no abridgments, adaptations, or segmented texts; (2) no teacher-sponsored, whole-class assignments; and (3) no direct instruction in isolated skill sequences' (p. 24). The other classifications, 'specific, broad, and eclectic', were applied to research studies that failed to meet these criteria in different ways. Jeynes and Littell incorporated multiple studies on language experience, including several of the same studies from the 1960s that were included in Stahl and Miller's (1989) meta-analysis, despite Stahl, McKenna, and Pagnucco's (1994) earlier warning that equating this with whole language was anachronistic.

Jeynes and Littell (2000) reported negative effect sizes for 'specific, broad, and eclectic' whole language studies – but found a high, positive effect size of 0.83 for the two studies classified as 'pure' whole language. One of these studies, 'A Comparison of Inner-City Children's Interpretations of Reading and Writing Instruction in the Early Grades in Skills-Based and Whole Language Classrooms' (Dahl & Freppon, 1994, p. 64), provides an example of the kind of study that genuinely contributes towards an effective comparison of whole language and rival approaches to early literacy. The researchers compared 'learner interpretations of beginning reading and writing instruction in skills-based and whole language inner-city classrooms across kindergarten and first grade' (p. 1), with students in eight classrooms, containing a majority of children from urban families with low income levels, in two American Midwest cities observed twice weekly over two years.

Dahl and Freppon (1999) explained:

The skills-based kindergartens included traditional reading readiness programs with extensive emphasis on letter/sound relations; the first-grade programs used a newly adopted traditional basal program with ancillary workbooks and dittos provided by the central administration. [...] Teachers followed the skill sequence in the basal program and met deadlines for unit completion established by the district. (p. 15)

By contrast:

The whole language classrooms utilized extended periods of self-selected independent reading and writing, and teachers worked with individual learners or small groups. The reading materials included a wide variety of children's literature and extensive classroom libraries. Instruction in first grade was carried out with whole-group sessions using extended storybook reading and included teacher demonstrations of reading strategies and skills. The writing program embraced writing workshop routines and used children's literature to suggest story themes and evoke topics. (Dahl & Freppon, 1994, p. 16)

As well as assessing quantitative data accumulated across the two year study, the researchers audio-taped their twice weekly classroom visits. This facilitated important research that addressed Goodman's (1989) warning about behavioural psychology-based approaches to literacy research that only examine post-test skills gains. Dahl and Freppon (1994) stated:

Particular attention was paid to learner statements and actions that indicated evolving hypotheses about reading and writing. [...] Thus, the outcome of each observation was an extended set of field notes in which transcripts of learner talk, oral reading samples, and learner actions were integrated. Copies of all learner papers (writing samples, ditto sheets) were also included. These elaborated accounts and artifacts were subsequently coded by the research team for learner behaviors and strategies, then analyzed for sense-making patterns. (pp. 19–20)

Dahl and Freppon (1994) concluded their 94-page report by noting that 'some findings, particularly those from quantitative measures, indicated a number of similarities in learning outcomes [between the whole language and skills-based classrooms]' (p. 63). In addition, however, Dahl and Freppon noted important qualitative differences in children's literacy behaviours. For example: 'Whole language learners generated significantly more syntactic and lexical features of story language and they experienced extended exposure to and interaction with storybooks' and:

Learners in whole language classrooms expressed extensive interest in themselves as literacy learners. Moreover, their talk and actions revealed an understanding of their strengths and weaknesses as readers

and writers. The linked patterns of sense of self as reader/writer and persistence indicated the establishment of a 'disposition for learning' and provided evidence of learner ownership and a positive attitude toward literacy. In the skills-based study these two patterns were evident only among the most proficient readers and writers. (p. 64)

Jeynes and Littell (2000) acknowledged that the positive effect sizes in what they classified as 'pure' whole language research studies, 'could indicate that when educators implement a clearly defined program of whole language, student performance improves' (p. 31).

Krashen's critique and the question of author bias

As with the Stahl and Miller (1989) meta-analysis on language experience, *Visible Learning* makes no reference to the secondary literature on the Jeynes and Little (2000) meta-analysis. This includes consideration of a critique by Krashen (2002), which raised a number of sharp objections to the Jeynes and Littell meta-analysis, including its definition of whole language, its inclusion of language experience research studies, its failure to take into account the time that students spent actually reading in each research study, and an inconsistent intervention benchmark of reading comprehension.

In addition, he examined several of the research studies included in the meta-analysis and calculated strikingly different effect sizes from those reported by Jeynes and Littell. Krashen (2002) explained:

I have argued that results were inaccurately reported in Carline and Hoffman (1976) and Lamb (1972) and that the direction of the effect size should be reversed in Harris and Serwer (Harris, Serwer & Gold, 1966). If my interpretations are correct, effect sizes for the still-eligible studies should be changed: for Hoffman, from -0.23 to +0.68; for Lamb, from -0.75 to +0.44; for Harris and Serwer, from -0.51 to 0.18, for Manning et al., from 1.21 to 1.97. (p. 39)

Krashen (2002) concluded:

Jeynes and Littell's conclusion was that although 'pure' whole language students did well, basal/skill groups were in general a winner over whole language, with a mean effect size of -0.65 in favor of the comparison groups. My conclusion is nearly exactly the opposite. [...] What is clear is that Jeynes and Littell's interpretation of the research is not the only possible one. (p. 40)

These conflicting calculations ought to have led Hattie to conduct his own review of the different effect sizes calculated by Jeynes and Littell. The scholarly record of the respective researchers would be one consideration potentially informing such a review. Critics of meta-analysis in medical science have discussed the potential impact of different forms of bias. LeLorier et al. (1997,

p. 536) noted, for example, 'Pooled results [in meta-analyses] incorporate the biases of individual studies and embody new sources of bias, mostly because of the selection of studies and the inevitable heterogeneity among them'. Hattie himself acknowledged the importance of the 'lens' through which different researchers examine evidence on a given area. It would, then, be remiss of anyone incorporating an extremely unusual reported effect size, such as $d = -0.65$ for whole language, to not consider the question of potential author bias.

There is an extensively documented hostility towards whole language teaching from American evangelical Christian groups and the Republican rightwing. Weaver and Brinkley (1998), for example, noted that in the US:

The primary push for intensive, systematic phonics comes originally – especially – from the religious right and more generally the far right, which currently wields considerable influence and power in our national and state governments. Phyllis Schlafly, for instance, is urging members of her Eagle Forum to get elected officials to propose pro-phonics legislation.' (p. 128)

Evangelical-Republican organisations led the 1990s backlash against whole language, culminating in the Bush administration's 2001 *No Child Left Behind* Act that prohibited federal funding for school districts that failed to promote synthetic phonics programs.

This history is relevant given that William H. Jeynes, lead researcher of the Jeynes and Littell meta-analysis, is an evangelist preacher and founder of God's Love Ministries. He also has close ties with the Republican Party and has said that he was among a shortlist of twelve candidates for Secretary of Education considered by President Donald Trump (Calvary Church of the Islands, 2017). Jeynes began academic work, he has explained, because:

I came to the conclusion that while ministers definitely make an impact, so do public schools. Public schools have our kids five days a week. Those kids are attending church, at most, three times a week. So the Lord laid it on my heart to go into a two-pronged ministry, both in churches and the academic sphere. (Kennedy, 2015)

Jeynes's publications (eight of which are used in *Visible Learning*) include: 'A Meta-Analysis on the Effects and Contributions of Public, Public Charter, and Religious Schools on Student Outcomes', 'The Impact of Parental Remarriage on Children: A Meta-Analysis', and 'The Relationship Between Biblical Literacy, Academic Achievement, and School Behavior Among Christian- and Public-School Students'. Before accepting Jeynes's -0.65 effect size for whole language, Hattie ought to have considered the possibility of whether author bias could have played any role in the

selection and interpretation of the research studies that comprised the Jeynes and Littell meta-analysis.

When a whole language meta-analysis reports a positive effect size

We have seen how *Visible Learning* accepted at apparent face value the Jeynes and Littell meta-analysis, and its finding of $d = -0.65$. Hattie treated very differently the Gee (1995) meta-analysis, and rejected its finding of $d = 0.65$.

We recall that the rival positive and negative 0.65 meta-analyses for whole language are not statistically equivalent. The average effect size of the 800+ meta-analyses included in Hattie's *Visible Learning* was 0.4. Whereas the Jeynes and Littell -0.65 was an extreme outlier, the Gee 0.65 was only somewhat higher than the overall mean. Given this, it is unusual that the negative 0.65 finding appears to have been uncritically accepted, whereas the positive 0.65 finding was subjected to careful scrutiny.

Hattie (2009) explained:

The average effect from the Gee study has an overall positive effect compared to the average from Jeynes and Littell (2000), which has an overall negative effect. There was much overlap in the studies used in these two meta-analyses and the difference is a function of how the authors classified some key studies, and the coding of what constituted whole language. On the latter, Gee included programs with systematic phonics and word study as whole language programs, and thus had what he considered a more 'balanced' program of both sets of skills. (p. 137)

On this basis, Hattie removed three research studies from the Gee meta-analysis and recalculated the effect size, changing it from 0.65 to zero, thereby accounting for the 'remarkable divergence' in whole language meta-analyses.

No other meta-analysis in *Visible Learning* was revised in this way. Hattie rejected several meta-analyses on learning styles, and he noted that an effect size for charter schools is lowered once low quality studies are excluded from meta-analysis, but this merely repeated a finding within a meta-analysis on the impact of charter schools. Out of more than 800 meta-analyses examined in *Visible Learning*, just one – the meta-analysis that found a positive impact for whole language teaching – had its effect size revised by Hattie through the selective elimination of certain research studies. It is difficult to explain this on any other grounds other than an evident prejudice against whole language. Hattie, it appears, has marshalled the evidence in a definite manner, accepting the dubious evidence against whole language presented by Jeynes and Littell, while selectively sifting data when whole language appears in a positive light, as in Gee (1995).

For at least one of the three research studies that Hattie excludes from Gee (1995), it is doubtful whether there are legitimate grounds for challenging its relevance as evidence related to whole language. Hattie states: 'Trachtenburg and Ferruggia's (1989) study, also classified by Gee as whole language, used various strategies including word lists, letter-sound naming, and decoding techniques' (p. 138).

On this basis, *Visible Learning* proposed to exclude the study, which had a reported effect size for whole language $d = 1.7$. We see again the importance of due consideration for a definition of whole language. Hattie appears to have worked with an implicit definition that precluded the possibility of teaching specific literacy skills through a whole language approach. Whole language theorists, however, have written numerous books and documents detailing instructional practices for phonics, decoding techniques, spelling; and other specific skills within a whole language framework (see, for example, Hornsby & Powell, 1993 and Hornsby & Wilson, 2011).

After explaining their interest in exploring 'the power of a whole language, shared book experience' (p. 284), Trachtenburg and Ferruggia (1998) explained: 'Many teachers embrace the philosophy of a literature based approach but are reluctant to attempt it for fear of neglecting requisite skillwork. This article dispels such fears by illustrating how skill instruction can flow naturally and logically from a literature base' (p. 285). Focusing on the use of big books, the researchers 'asked students to choose one word each day from the Big Book that they wished to learn' (p. 286). They also published a story from a student retell, using the students' language to subsequently focus on specific letter-sound relationships. None of this is at odds with whole language. Hattie's 'lens' on the evidence nevertheless led him to exclude the Trachtenburg and Ferruggia study, as part of what appears as a concerted drive in *Visible Learning* to reduce to zero the overall positive effect size for whole language found in the Gee meta-analysis.

Visible Learning's presentation of literacy research suggests that in addition to working with an implicit definition of whole language, there is a broader, unstated definition or understanding of what reading itself consists of. At different points in the book, without explanation, 'comprehension' is distinguished from 'reading', for example when Hattie (2009, p. 222) reports on separate effect sizes for computer-based instruction of reading and for computer-based instruction of comprehension. He also, we recall, concluded that whole language programs 'have negligible effects on learning to read – be it on word recognition or on comprehension' (Hattie, 2009, p. 138). At another point

in *Visible Learning*, it is asserted (without reference to any meta-analysis or research study) that 'phonics often precedes comprehension, and placing too much emphasis on the latter before the former is learnt (as is typical of many whole language programs) is not effective' (Hattie, 2009, p. 160).

This all points to a possible influence within *Visible Learning* of the so-called 'simple view' of reading, which argues that 'learning to decode is tantamount to learning to read' (p. 6) and that 'reading ability should be predictable from a measure of decoding ability (e.g., the ability to pronounce pseudo-words) and a measure of listening comprehension' (Gough & Tunmer, 1986, p. 7). This is a perspective that from the outset was explicitly directed against whole language theory, which insists that reading *is* comprehension (see, e.g., Cambourne 1988). The 'simple view' was notably promoted through the British government's Rose Report into the teaching of reading (Rose, 2006) that preceded the imposition of mandatory synthetic phonics instruction in that country. The 'simple view' remains highly contested; Proctor (2006, p. 1) noted that many researchers argue it is 'far too simple to make a meaningful difference in understanding the vast complexities that individual learners bring to the reading process'. Harrison (2010, p. 207) noted that the 'simple view' has 'become associated with the "first, fast and only" model of phonics, and a technicist view of the pedagogy of reading'.

Conclusions

A review of *Visible Learning*'s 'meta-meta-analysis' on whole language demonstrates the impossibility of meaningfully engaging with literacy research solely on the basis of examining attributed effect sizes in student achievement. Research in the teaching of reading to young children is complex and contested; in stripping out all the nuances, and ignoring all secondary literature on the methodology and conclusions of different meta-analyses, Hattie's assigned effect sizes for different aspects of literacy learning cannot be legitimately used to present direct instruction and synthetic phonics as 'evidence based'. Attempting to orient pedagogical practice through *Visible Learning*'s reported effect sizes is ill advised, while basing public policy decisions on them is irresponsible.

The promise that meta-analysis could resolve 'past difficulties associated with summarising and comparing all the diverse types of evidence about what works in classrooms' (Hattie, 2009, p. 3) has proved a false one, ultimately resting upon a technicist pedagogical 'story'. As noted by Wrigley (2018, p. 13): 'The sloganistic 'what works' reflects a neoliberal demand to extract

maximum efficiency from education, while marginalising the qualitative and political dimensions of human formation.'

Bloome, Harris and Ludlam (1991, p. 15) stated that literacy constitutes a complex web of sociocultural activities, with the 'overarching implication of understanding reading and writing as social and cultural practices is recognition that pedagogy is always political (involving relationships of power)'. Complex power relations involving rival political, social, and economic forces are evident throughout the history of literacy instruction in English-speaking countries. The emergence of the whole language movement in the 1970s was the product not only of developments in literacy theory and research but also broader political developments, including the civil rights and other social movements in the US during this period (see Y. Goodman, 1989; Ryan & Goodman, 2016). Within education, these developments lent greater urgency to the development of child-centred literacy teaching that recognised and respected individual cultural and linguistic differences (Y. Goodman, 1989, p. 121).

Visible Learning can likewise be situated in its specific historico-political context. Wolfe and Poynor (2001, p. 15) challenged the popular metaphor of pendulum swings within education policy to argue instead 'that the rise and fall of this theoretical and pedagogical movement [whole language] more closely resembles a sociopolitical movement inextricably linked to conflicts over hegemonic control'. Since the 1980s, the dominance of neoliberal politics internationally has been accompanied by a revival of the science/technology view of literacy, promoted by different coalitions of 'free market', corporate-funded think tanks, lobby groups, and edu-businesses. As noted by Soler (2016, p. 431): 'Synthetic-phonics approaches to early literacy fit a neoconservative traditional view of knowledge and the curriculum. They also fulfil a neoliberal agenda for rationality within a centrally controlled model of top-down delivery'. False claims that cognitive science has developed a definitive, 'evidence based', synthetic phonics based method of literacy instruction, with meta-analysis promoted as part of this, forms an important part of the associated technicist rhetoric (see Wrigley, 2018).

Hattie makes no bones in *Visible Learning* about his enthusiasm for direct instruction and hostility towards what he variously describes as 'Rousseian inspired methods' (p. 258) and 'facilitative and inquiry methods' that are driven by a 'constructivist fad' (p. 243). *Visible Learning* concludes (pp. 244–245) with Hattie promoting his ideal 'teacher as activator', based on direct instruction, as opposed to 'teacher as

facilitator', under which label he groups inquiry-based teaching, smaller class sizes, problem-based learning, and whole language.

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